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Mathematical modeling of the influence of non-catalytic dissociation / recombination of water molecules in the desalination channel on electric convection

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Abstract. The article formulates a two-dimensional mathematical model of non-stationary transport of 1: 1 electrolyte in a potentiodynamic mode, taking into account electroconvection and non-catalytic dissociation / recombination reaction of water molecules in electromembrane systems, which are considered as the desalting channel of an electrodialysis device. The model is described by a system of coupled Navier-Stokes and Nernst-Planck-Poisson equations taking into account the electric force and physically justified boundary conditions. The article establishes the basic laws of mass transport, taking into account the dissociation / recombination of water molecules. It was shown for the first time that a double electric layer of hydrogen and hydroxyl ions arises in the recombination region. It is shown that between the region of recombination and quasi-equilibrium regions of space charge there are regions of electroneutrality and equilibrium with an almost linear distribution of concentrations. It was found that even under prelimiting, but close enough to the limiting current, modes, non-catalytic dissociation of water molecules in the quasi-equilibrium region of space charge occurs so intensely that the concentration of hydrogen and hydroxyl ions becomes comparable to the concentration of potassium and chlorine ions. At overlimiting current densities, due to the appearance of an extended space charge region and intense dissociation of water molecules in this region, as well as an increase in the electric double layer in the recombination region, the space charge and the dissociation / recombination reaction of water molecules significantly affect each other. In turn, this has a decisive effect on electroconvection and, accordingly, on the transport of salt ions.

1. Introduction

The dissociation of water molecules in systems with ion-exchange membranes was first noted in works by Frilett [1], Kressman and Tye [2] and further studied in works [3-6]. The dissociation reactions of water molecules and their reverse recombination are important, since the appearance of new charge carriers can lead to a decrease or even disappearance of electroconvection, the main mechanism of the overlimiting transport of salt ions. There are two different mechanisms for the dissociation of water in electromembrane systems, namely catalytic and non-catalytic. According to the catalytic mechanism of water dissociation on ion-exchange membranes, and ions can arise in the reactions of proton transport between charged groups in the membrane and water molecules [7, 8]. It was found that water dissociation is more pronounced on anion exchange membranes (AOM) and not



so significant on cation exchange membranes (COM) [8]. In a number of works [9, 10], we showed that in the extended space charge region, the electric field strength reaches such high values that it pulls hydrogen and hydroxyl ions in this region and a noncatalytic reaction of dissociation of water molecules occurs at the maximum possible rate. We also in work [11], using a one-dimensional model, showed the emergence of a double electric layer inside the section of the desalination channel.

In this work, a two-dimensional mathematical model is formulated and, on the basis of its numerical analysis, the mutual influence of the emergence and development of space charge, the reactions of dissociation / recombination of water molecules and electroconvection and their joint effect on the transport of salt ions are investigated.

Comment. Speaking about H^+ ions, a simplification is made, since they are absent in water. The proton "lost" by the water molecule is attached to another molecule to form the hydronium ion H_3O^+ , and the water dissociation reaction is written as follows.



Hydronium ion, in turn, can combine with other water molecules to form hydrated ions, for example, $H_5O_2^+$, $H_7O_3^+$, $H_9O_4^+$.

A two-dimensional diagram of the desalting channel of an device (Figure 1a) for simulating the unsteady transport of 1:1 electrolyte in the potentiodynamic mode, taking into account electroconvection and the noncatalytic reaction of dissociation / recombination of water molecules in electromembrane systems (Figure 1b) is shown in Figure 1. The flow channel of the desalination of the electrodialysis apparatus is investigated. Let us assume that the Oy axis passes through the anion-exchange membrane, and the Ox axis perpendicular to it is directed towards the cation-exchange membrane. Let the channel width be equal H , and L is the length. An anion-exchange membrane is located at a point $x=0$, a cation-exchange membrane is located at a point $x=H$. A dilute NaCl solution is fed to the entrance to the desalination channel located at $y=0$. The exit from the channel is located at $y=L$.

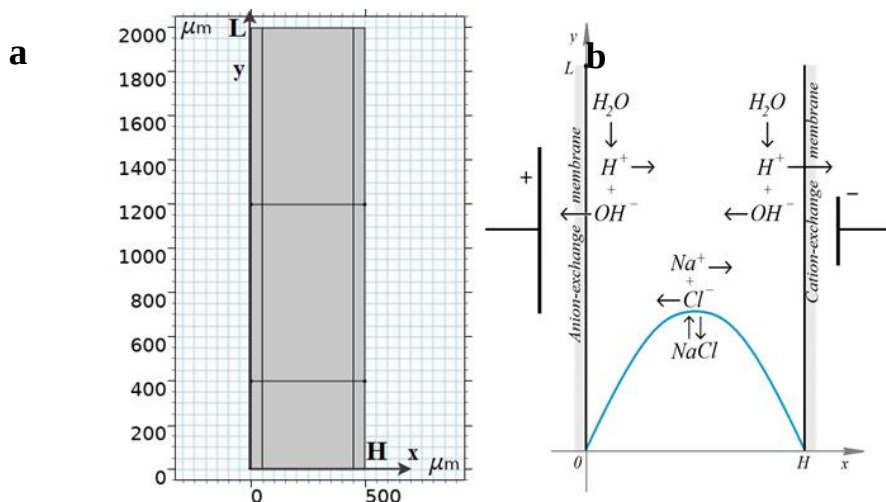


Figure 1. Diagram of the desalination channel and simulated phenomena

2. Mathematical model

2.1. System of Equations

The unsteady transport of a binary electrolyte taking into account the reaction of dissociation / recombination of water molecules and electroconvection in a desalination channel electrodialysis

device is described by a coupled system of two-dimensional system of coupled Navier-Stokes and Nernst-Planck-Poisson equations, taking into account the electric force (dimensional view and vector notation) (1-6). The Nernst-Planck-Poisson and Navier-Stokes equations are the expression of the general conservation laws, therefore, the model of the transport of salt ions, taking into account electroconvection, is determined, along with the formula for the electric force, in the Navier-Stokes equation, first of all, by the boundary conditions. One of the variants of the boundary conditions is described below. In the potentiodynamic mode, the surfaces of the ion-exchange membranes are considered homogeneous and equipotential, and the potential drop is set.

$$\frac{\partial C_i}{\partial t} = -\text{div } \vec{j}_i + R_i, \quad i = 1, \dots, 4 \quad (1)$$

$$\vec{j}_i = -z_i \frac{F}{RT_0} D_i C_i \nabla \varphi - D_i \nabla C_i + C_i \vec{V}, \quad i = 1, \dots, 4 \quad (2)$$

$$\Delta \varphi = -\frac{F}{\varepsilon_r} (z_1 C_1 + z_2 C_2 + z_3 C_3 + z_4 C_4) \quad (3)$$

$$R_1 = R_2 = 0, \quad R_3 = R_4 = k_d C_{H_2O} - k_r C_3 C_4 = k_r (k_w - C_3 C_4) \quad (4)$$

$$\vec{I}_c = F \sum_{i=1}^4 z_i \vec{j}_i, \quad z_1 = 1, z_2 = -1, z_3 = 1, z_4 = -1 \quad (5)$$

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \nabla) \vec{V} = -\frac{1}{\rho_0} \nabla P + \nu \Delta \vec{V} + \frac{1}{\rho_0} \vec{f}, \quad \text{div}(\vec{V}) = 0 \quad (6)$$

where Δ is the Laplace operator, ∇ is the gradient, $\vec{V}$ is the flow rate of the solution, ρ_0 is the characteristic density of the solution, P is the pressure, $\vec{j}_i, C_i, i = \overline{1,4}$ are the flows and concentrations of cations, anions, hydrogen and hydroxyl in the solution, respectively, $z_i, i = \overline{1,4}$ are the charge numbers of cations and anions, $\vec{I}$ is the current density, $D_i, i = \overline{1,4}$ are the coefficients diffusion of cations and anions, hydrogen and hydroxyl, respectively, φ is the potential of the electric field, ε_r is the dielectric constant of the electrolyte, F is the Faraday constant, R is the gas constant, T is the absolute temperature, t is the time, ν is the coefficient of kinematic viscosity, $\vec{f}$ is the density of the electric force: $\vec{f} = \rho \vec{E} = -\varepsilon_r \Delta \varphi \vec{E} = \varepsilon_r \Delta \varphi \nabla \varphi = \varepsilon_r \vec{E} \text{div} \vec{E}$, where $\rho = F \sum_i z_i C_i, i = \overline{1,4}$ is the distribution

density space charge, and $\vec{E} = -\nabla \varphi$ is electric field strength. The unknown functions are functions of time and x, y coordinates.

2.2. Boundary conditions

The surfaces of the ion exchange membrane are considered to be ideally selective, i.e. impermeable to co-ions, the boundary concentration of counter-ions is equal to the fixed charge inside the membrane. The adhesion condition is set for the velocity on the membrane surface.

Near the anion exchange membrane $x = 0$:

$$-\vec{n} \cdot \left(-\frac{F}{RT} C_1 D_1 \nabla \varphi - D_1 \nabla C_1 \right) \Big|_{x=0} = 0, \quad (7)$$

$$C_2(t, 0, y) = C_{2a}, \quad (8)$$

$$-\vec{n} \cdot \left(-\frac{F}{RT} C_3 D_3 \nabla \varphi - D_3 \nabla C_3 \right) \Big|_{x=0} = j_{3a} \quad (9)$$

$$-\vec{n} \cdot \nabla C_4(t, 0) = 0 \quad (10)$$

$$-\vec{n} \cdot \vec{v}(t, 0, y) = 0 \quad (11)$$

$$\varphi(t, 0, y) = \Delta_r \varphi \cdot t \quad (12)$$

Near the anion exchange membrane $x = H$, where H is the channel section width:

$$C_1(t, H, y) = C_{1k} \quad (13)$$

$$-\vec{n} \cdot \left(\frac{F}{RT} C_2 D_2 \nabla \varphi - D_2 \nabla C_2 \right) \Big|_{x=H} = 0 \quad (14)$$

$$-\vec{n} \cdot \nabla C_3(t, H, y) = 0 \quad (15)$$

$$-\vec{n} \cdot \left(\frac{F}{RT} C_4 D_4 \nabla \varphi - D_4 \nabla C_4 \right) \Big|_{x=H} = j_{4k} \quad (16)$$

$$-\vec{n} \cdot \vec{v}(t, H, y) = 0 \quad (17)$$

$$\varphi(t, H, y) = 0 \quad (18)$$

At the entrance to the region of ion concentration and the potential distribution are assumed to be given, so that the condition of electroneutrality is satisfied at the entrance, i.e. [1]:

$$C_i(x, 0, t) = C_{i,0}, i = 1, \dots, 4,$$

$$z_1 C_1(x, 0, t) + z_2 C_2(x, 0, t) + z_3 C_3(x, 0, t) + z_4 C_4(x, 0, t) = 0$$

Depending on the objectives of the study, the distribution of the potential at the input will be set in different ways, or by a linear function (18), or by (19). With a constant distribution of concentrations, the formula (18) follows from the fulfillment of Ohm's law. Thus, the formula (18) is consistent with Ohm's law and constant concentrations at the entrance to the channel [12]. For a symmetric electrolyte with the same diffusion coefficients, we obtain condition (19).

$$\varphi(x, y, t) = d_\varphi x / H \quad (19)$$

$$\frac{\partial \varphi(x, 0, t)}{\partial y} = 0 \quad (20)$$

Depending on the objectives of the study, $\vec{V}$ the flow rate of the solution at the inlet will be considered constant, or distributed over the Poiseuille parabola (20), where V_0 is the average flow rate of the solution.

$$V_x = 0, V_y = 6V_0 \frac{x}{H} \left(1 - \frac{x}{H}\right) \quad (21)$$

At the exit from the region $y=L, x \in [0, H], t \geq 0$, the concentration condition is used for the ion flux, which means that salt ions are removed from the desalination channel only due to the solution flow. A condition $-\vec{n} \cdot \nabla \varphi = 0$ is set for the potential. Since unsteady vortices can arise in the problem, the boundary conditions for the velocity at the exit from the channel may contain errors. Two methods have been used to overcome this problem. The first is lengthening the channel and extracting data at some distance before exiting. In this case, we can assume that the outlet flow is a Poiseuille flow. However, this often leads to a significant increase in the calculation time. Also, this method should be used with caution in the presence of electroconvection at the exit from the channel. In the second method of modeling, the boundary is considered open and the normal stress on it is equal to zero, which will allow the vortices to flow freely from the area under study. However, in numerical experiments, this condition often causes difficulties in finding the initial approximation in Newton's method when using the finite element method.

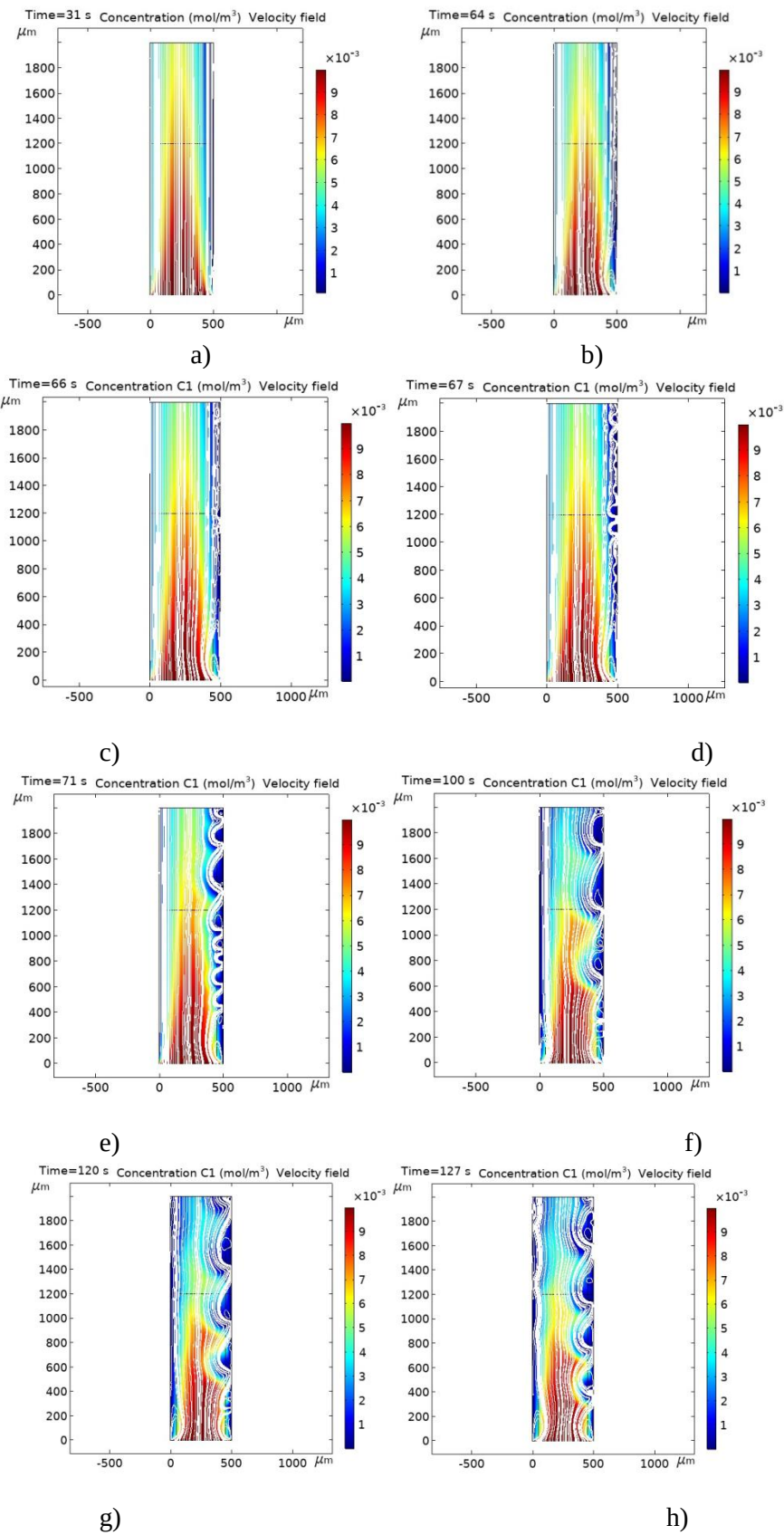
The initial conditions $t=0$ are set consistent with the rest of the conditions. In a numerical solution, the inconsistency of the initial and boundary conditions is smoothed out rather quickly. The conditions for the constant concentration at the inlet and the condition for the ideal selectivity of ion-exchange membranes used in this study, although not agreed at the corner points of the channel inlet, but, as numerical calculations show, smooth out rather quickly and do not lead to any artifacts. Naturally, for this it is necessary in the numerical solution to refine the mesh at the corner entry points.

3. Results of numerical experiments

The finite element method implemented in the Comsol Multiphysics 5.5 environment was used to numerically solve the boundary value problem.

Let us first consider the flow of a solution taking into account electroconvection, but without taking into account the influence of the dissociation / recombination reaction of water molecules. As can be seen from Fig. 1, electroconvection begins with the formation of a "stationary vortex" at the entrance to the channel near the cation-exchange membrane rather quickly (with our data in Fig. 1a at time 31 s. Over time, the "stationary vortex" begins to gradually increase, expanding both across and in the first. At some moments of time (Fig. 1b at 64 s) a small electroconvective vortex breaks away from the electroconvective "stationary vortex" and moves downstream, gradually decaying (in Fig. 1c, this vortex decays at the time 66 s) at this. At the same time, the stationary vortex expands significantly downstream, a disturbance develops gradually downstream, and at a certain moment of time (Fig. 1d at $t = 67$ s) vortex formation begins along the entire cation-exchange membrane. membranes, since, starting from this moment in time, developed vortices and vortex clusters appear, with holding several vortices inside themselves, bifurcation and dynamic chaos of vortices (see. for example Fig. 1f $t = 100$ s). Subsequently, according to the same scheme, electroconvection begins and develops at the anion-exchange membrane (Fig. 1e-1j at 72s, 100s, 120s, 127s, 130s, 142s).

The above-described scheme of electroconvection (in the absence of a dissociation / recombination reaction of water molecules) substantially depends on the sweep rate of the potential jump. If this speed is sufficiently low, then electroconvection will develop according to a different scheme, namely, vortex formation at the ion-exchange membranes begins at the exit from the channel, while there is no "stationary vortex" at the entrance to the channel [13-10].



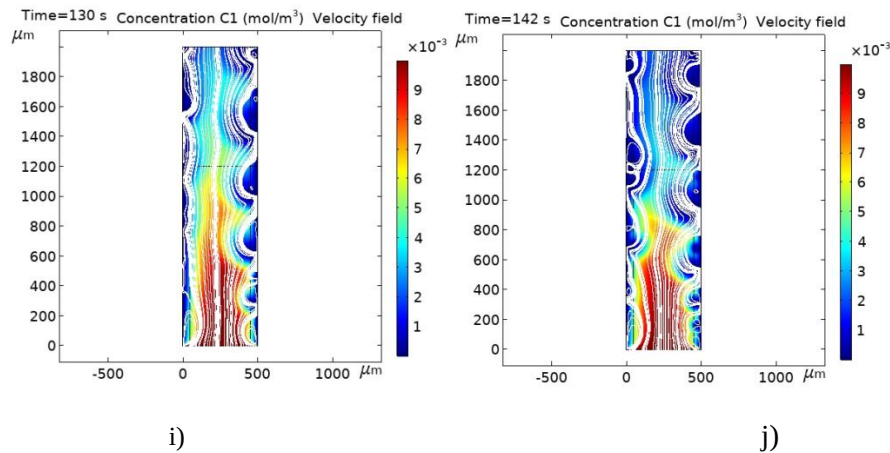
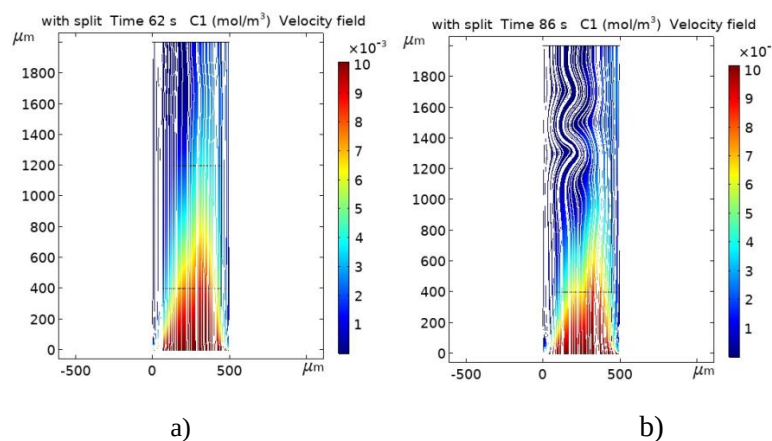
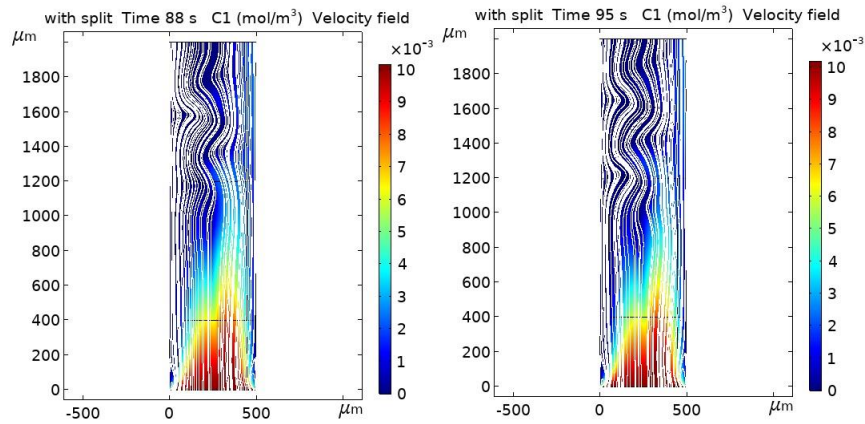


Figure 2. Streamlines of the solution taking into account electroconvection, but without taking into account the dissociation / recombination reaction of water molecules

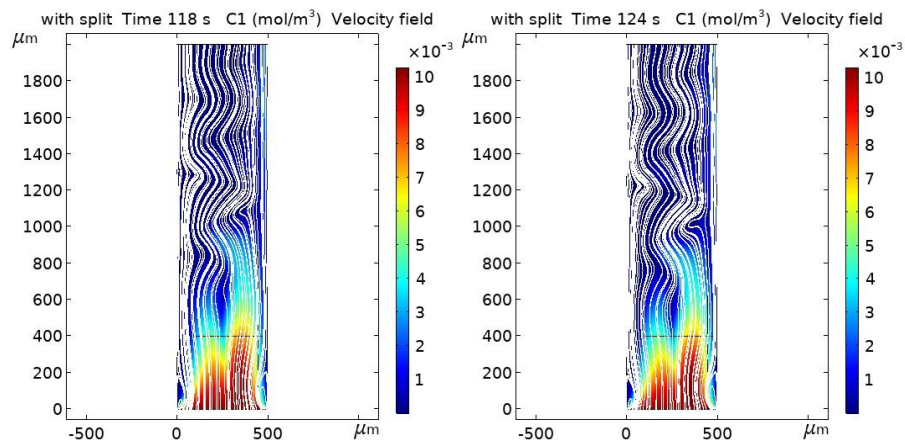
Let us now consider the formation and development of electroconvection taking into account the reaction of dissociation / recombination of water molecules. This reaction significantly alters both the formation and development of electroconvection (see Figure 2). The formation of an electroconvective vortex begins approximately in the middle part of the channel (shifted towards the anion-exchange membrane) closer to the end of the channel (this occurs approximately at $t = 62$ c (Fig. 2a)). This electroconvective vortex is similar to the stationary vortex described above. Over time, it gradually expands upstream. At some point in time (Fig. 2b at 86c), the streamlines of the solution begin to bend strongly, starting from the anion-exchange membrane more than in half of the channel, excluding almost a third of the channel adjacent to the channel entrance. At some point in time (Fig. 2c at $t = 88$ c), the formation of electroconvective vortices begins at the anion-exchange membrane, which flow downstream, but each new vortex is formed closer and closer to the channel entrance, however, not reaching the channel entrance. At this time, the cation-exchange membrane has no electric convection, however, at some point in time (Fig. 2d 95c), an electroconvective vortex appears practically in the middle part of the desalination channel, which then flows downstream, gradually some of these vortices are formed closer to the cation-exchange membrane, and the other part, remaining approximately in the middle of the channel, forms closer and closer to the channel entrance (Fig. 2e 118c and 2f 124c). Further, these processes become more and more evident (in Figure 2g, h 134c and 143c). Such a complex process of formation and development of electroconvective vortices is associated with a complex structure of the space charge region caused by the influence of the dissociation / recombination reaction of water molecules on the space charge distribution in the desalination channel (in Figure 2i, j).





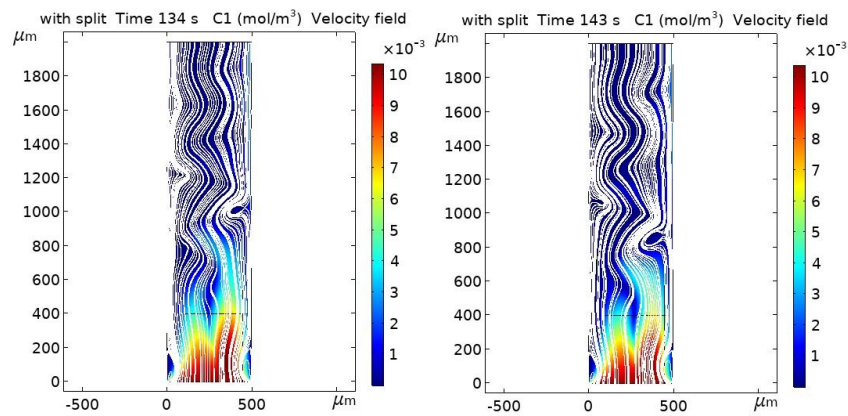
c)

d)



e)

f)



g)

h)

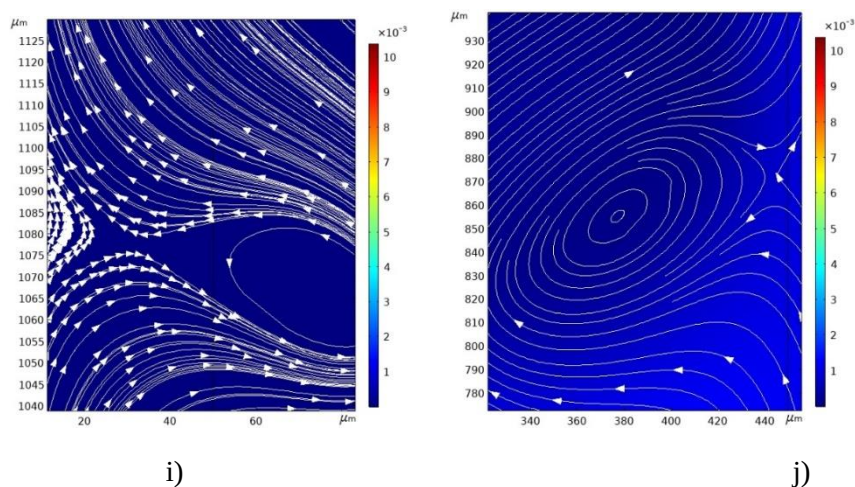


Figure 3. The streamlines of the solution taking into account electroconvection and the reaction of dissociation / recombination of water molecules at different times and an increase in two vortices at $t = 143s$ with the coordinates of the centers i): (0.8mm; 1.065mm) and j) :(0.38mm; 0.855mm). The arrows show the direction of movement of the solution.

4. Conclusion

In this article, using a mathematical model, for the first time, the unsteady transfer of ions of a binary salt in the desalination channel is analyzed taking into account the reaction of dissociation / recombination of water molecules. The main regularities have been established, namely, it has been shown that the cross-section of the desalination channel consists of narrow quasi-equilibrium regions of space charge adjacent to ion-exchange membranes. In the central part, there is a fairly extensive region of recombination, where recombination dominates over dissociation. It was shown for the first time that a double electric layer of hydrogen and hydroxyl ions arises in the recombination region. It is shown that between the region of recombination and quasi-equilibrium regions of space charge there are regions of electroneutrality and equilibrium with an almost linear distribution of concentrations. It was found that even under prelimiting, but close enough to the limiting current, modes, non-catalytic dissociation of water molecules in the quasi-equilibrium region of space charge occurs so intensely that the concentration of hydrogen and hydroxyl ions becomes comparable to the concentration of potassium and chlorine ions. At overlimiting current densities, due to the appearance of an extended space charge region and intense dissociation of water molecules in this region, as well as an increase in the electric double layer in the recombination region, the space charge and the dissociation / recombination reaction of water molecules significantly affect each other. In turn, this has a decisive effect on electroconvection and, accordingly, on the transport of salt ions.

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